

Descriptive Statistics Project

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Description: Number of Summer Gold Olympic Medals from 1896-2008 by these
listed countries: Afghanistan, Algeria, Argentina, Armenia, Australasia, Austria,
Azerbaijan, Bahama, Belarus, Belgium.

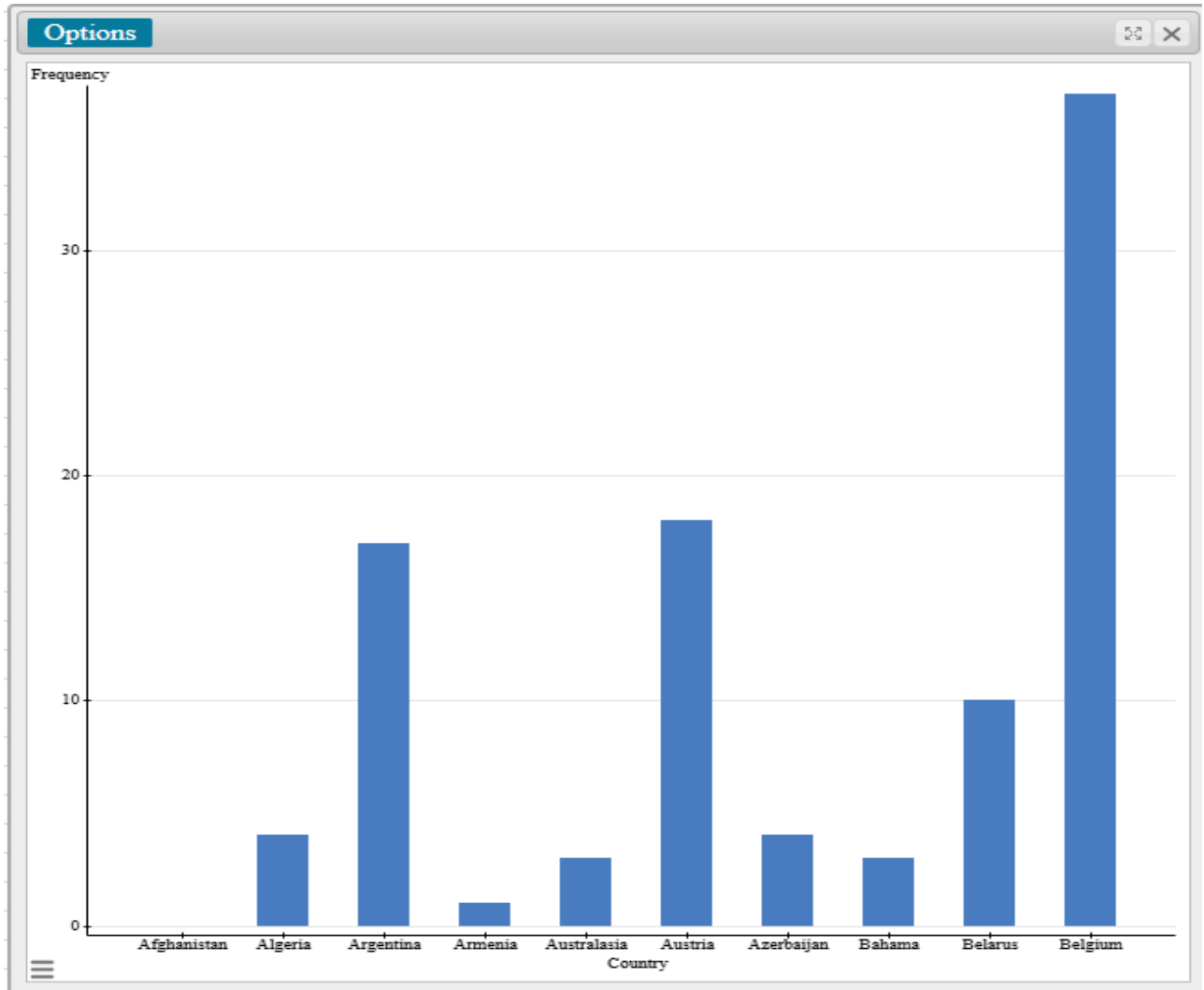
Direct Link: <https://www.statcrunch.com/app/index.html?dataid=4973291>

Objectives:

I shall do the following objectives:

- (1.) Dataset Description: Describe the dataset.
- (2.) Data Presentation: Present the dataset using the histogram.
- (3.) Measures of Central Tendency: Calculate and interpret the measures of center.
- (4.) Measures of Variation: Calculate and interpret the measures of spread.
- (5.) Measures of Location: Calculate and interpret the measures of position.

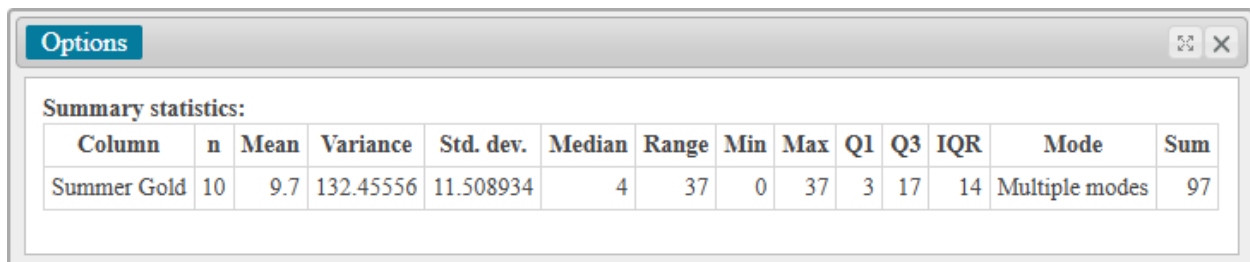
Data Presentation:



Country ▼	Summer Gold
Afghanistan	0
Algeria	4
Argentina	17
Armenia	1
Australasia	3
Austria	18
Azerbaijan	4
Bahama	3
Belarus	10
Belgium	37

Of the listed countries, Belgium has the highest number of Summer Gold Olympic Medals from 1896-2008 while Afghanistan has none.

Data Analysis:



The screenshot shows a software window titled "Options" with a close button. Inside, under the heading "Summary statistics:", there is a table with 13 columns: Column, n, Mean, Variance, Std. dev., Median, Range, Min, Max, Q1, Q3, IQR, Mode, and Sum. The data row for "Summer Gold" shows n=10, Mean=9.7, Variance=132.45556, Std. dev.=11.508934, Median=4, Range=37, Min=0, Max=37, Q1=3, Q3=17, IQR=14, Mode=Multiple modes, and Sum=97.

Column	n	Mean	Variance	Std. dev.	Median	Range	Min	Max	Q1	Q3	IQR	Mode	Sum
Summer Gold	10	9.7	132.45556	11.508934	4	37	0	37	3	17	14	Multiple modes	97

We shall use formulas, Pearson Statcrunch software, and Texas Instrument (TI-84 CE) calculator.

We shall begin with the measures of center. I shall round all final answers to **two decimal places as needed**.

Measures of Center:

These are: Mean, Median, Mode, and Midrange.

0, 1, 3, 3, 4, 4, 10, 17, 18, 37

Sample Size = 10

$$\text{Mean} = \frac{\text{Sum of data values}}{\text{Sample Size}} = \frac{97}{10} = 9.7$$

The average number of Summer Gold Olympic Medals from 1896-2008 by the listed countries is 9.7 which is approximately 10 gold medals.

Median: I shall arrange the dataset in ascending order.

0, 1, 3, 3, 4, 4, 10, 17, 18, 37

$$\text{Median} = \frac{4 + 4}{2} = 4$$

The median number of Summer Gold Olympic Medals from 1896-2008 by the listed countries is 4 medals.

Mode: There are two sets of two different countries with the same numbers of gold medals. Hence, the mode is 3 and 4.

Midrange:

Minimum: 0

Maximum: 37

$$\text{Midrange} = \frac{\text{minumun} + \text{maximum}}{2} = \frac{0 + 37}{2} = 18.5$$

The average of the minimum and the maximum number of Summer Gold Olympic Medals from 1896-2008 by the listed countries is 18.5 which is approximately 19 medals.

Measures of Spread:

These are the: Range, Variance, Standard Deviation, and Interquartile Range.

Let us calculate the range.

$$\text{Range} = \text{maximum} - \text{minimum}$$

$$\text{Range} = 37 - 0 = 37$$

The difference between the maximum value and the minimum value of Summer Gold Olympic Medals from 1896-2008 by the listed countries is 37 medals.

Variance and Standard Deviation:

I shall round all intermediate calculations to 3 decimal places as needed.

Summer Gold, x	$x - \text{Mean}$	$(x - \text{Mean})^2$
0	-9.7	94.09
4	-5.7	32.49
17	7.3	53.29
1	-8.7	75.69
3	-6.7	44.89
18	8.3	68.89
4	-5.7	32.49
3	-6.7	44.89
10	0.3	0.09
37	27.3	745.29

$$\Sigma(x - \text{Mean})^2 = 1192.1$$

$$\text{Variance} = \frac{\Sigma(x - \text{Mean})^2}{n - 1} = \frac{1192.1}{9} = 132.4555556$$

$$\text{Standard Deviation} = \sqrt{\text{Variance}} = \sqrt{132.4555556} = 11.50893373$$

There is a considerable amount of spread in the number of Summer Gold Olympic Medals from 1896-2008 by the listed countries.

Interquartile Range, IQR:

The dataset arranged in ascending order is:

0, 1, 3, 3, 4, 4, 10, 17, 18, 37

$$n = 10$$

$$\frac{1}{4} * 10 = 2.5 \cong 3 \text{ (rounded up)}$$

Lower Quartile, Q_1 = 3rd position = 3 gold medals

$$\frac{3}{4} * 10 = 7.5 \cong 8 \text{ (rounded up)}$$

Upper Quartile, $Q_3 = 8th\ position = 17$ gold medals

$$IQR = Q_3 - Q_1 = 17 - 3 = 14$$

The variation in 50% of the number of Summer Gold Olympic Medals from 1896-2008 by the listed countries is 14 medals.

Measures of Position:

I shall use the five-number summary of data.

$$Minimum = 0$$

The minimum number of Summer Gold Olympic Medals from 1896-2008 by the listed countries is 0 medals.

$$\text{First Quartile, } Q_1 = 3$$

25% of the number of Summer Gold Olympic Medals from 1896-2008 by the listed countries is below 3 medals.

Middle Quartile, $Q_2 = 4$

50% of the number of Summer Gold Olympic Medals from 1896-2008 by the listed countries is below 4 medals.

Upper Quartile, $Q_3 = 17$

75% of the number of Summer Gold Olympic Medals from 1896-2008 by the listed countries is below 17 medals.

Maximum = 37

100% of the number of Summer Gold Olympic Medals from 1896-2008 by the listed countries is below 37 medals.

References (MLA 9):

Chukwuemeka, Samuel. "Descriptive Statistics." *Appspot.com*, 2015,
conferencepresentations.appspot.com/Projects/Statistics/DescriptiveStatistics.html.
Accessed 20 Nov. 2025.

"StatCrunch." *Statcrunch.com*, 2025,
www.statcrunch.com/app/index.html?dataid=4973291. Accessed 20 Nov. 2025.